Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_09

1. Summary of Key Deviations

The analysis of Run_09 reveals a complex and severe dual-fault scenario, involving both a major physical blockage and significant over-performance of the pump.

- **Flow_Bypass (Critical):** The most extreme symptom is the bypass flow, which goes into **saturation**, reading **+100.2%** higher than healthy. This indicates an enormous amount of fuel is being diverted from the engines.
- **Flow_Pump (Moderate):** The pump flow shows a very large positive **offset**, reading almost **+30%** higher than the healthy baseline. This indicates the pump is not only compensating for a blockage but is actively and significantly over-performing.
- **Flow_Engine1-4 (Moderate):** All four engine flows show a major, consistent **negative offset** of approximately **-26%**. This confirms that fuel is being severely restricted from reaching the combustion chamber.
- **Pressure Signals (Moderate):** System pressures (`p_FMU`, `p_Shut`) show a moderate **negative offset**, which is counter-intuitive given the blockage but likely caused by the extreme flow dynamics in the bypass loop.
- **Flow_PRV (Nominal):** There is no significant deviation on the `Flow_PRV` signal.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Fuel Metering Unit (FMU) & Main Fuel Pump**	**Dual Fault: Severe Blockage & Pump Over-performance**	This is the only hypothesis that fits all the evidence. A **severe blockage at the FMU** explains why all `Flow_Engine` readings are low (-26%) and why `Flow_Bypass` is saturated (+100%). Simultaneously, a separate fault is causing the **Main Fuel Pump to massively over-perform** (+30%). The combination of these two faults creates the observed scenario: the over-performing pump generates huge amounts of excess flow, and with the path to the engines blocked, almost all of this excess flow is forced through the bypass loop, saturating its sensor.

2	**Fuel Metering Unit (FMU)**	**Severe Partial Blockage**	A blockage at the FMU alone would explain the low `Flow_Engine` and high `Flow_Bypass`. However, it does not explain why `Flow_Pump` is reading +30% higher than normal. In a simple blockage case, the pump flow should be closer to nominal.
3	**Main Fuel Pump**	**Severe Over-performance**	Pump over-performance alone would cause both `Flow_Pump` and `Flow_Bypass` to be very high. However, it does not explain why all `Flow_Engine` signals are uniformly low, which proves there must also be a significant downstream blockage.
4	**Bypass Valve**	**Stuck Open**	A stuck-open bypass valve would cause high bypass flow but would lead to a *drop* in pump flow and system pressure, which contradicts the observed +30% pump flow.
5	**Multiple Sensor Failures**	**Various Errors**	It is extremely unlikely that the `Flow_Engine` sensors would all fail with a negative offset while the `Flow_Pump` and `Flow_Bypass` sensors fail with a large positive offset. The pattern is clearly indicative of a complex but physically consistent dual-fault event.

3. Most Probable Hypothesis

The most probable root cause is a **Dual Fault Scenario: (1) a Severe Blockage at the Fuel Metering Unit (FMU) AND (2) a Significant Over-performance of the Main Fuel Pump.**

Reasoning: No single fault can explain the full set of symptoms. The data presents two distinct and overlapping problems. The first is a classic downstream blockage, proven by the low engine flows and extremely high bypass flow. The second is a pump that is delivering far too much fuel (+30%). A simple blockage wouldn't cause the pump to over-perform this dramatically, and simple over-performance wouldn't cause all engine flows to drop. It is the combination of these two independent failures that creates this unique and severe diagnostic signature, where a wildly over-performing pump is forcing its excess flow into a bypass loop because its primary path to the engines is obstructed.